

SWYFT Record Twin, status at day 13 of 26

A right-of-way record map goes in. A checked, georeferenced object record comes out, on the LiDAR, with a queue of what could not be proven. This page says where that stands on 2026-09-23, twelve days before the demo.

DATE	2026-09-23
DEMO	2026-10-05, 25 min live
BRANCH	caltrans-spike, loop 11 (2026-09-26 04:40)
DEMO SHEET	R-10434.2, Presidio, SF-101
TEST SET	R-10434.1/.2/.3 (Presidio tile) + R-10741.1/.2/.3 (Marin tile) + gap tile (3DEP 2018) · bench also R-71.71, R-10258.1, R-71.70 · 1969 R-65.1..4
RUN	python spike/demo.py --fast · 245 s (regressed, see next steps) · --six 923 s

83% printed distances that match the drawn line 41/49 on R-10434.2 · was 44%	0.04 ft georeferencing RMS, 18 of 19 control points no hand-typed coordinates	44/44 table rows read clean from the CAD text 111/111 keyed cells match	1 figure closed by traverse from the printed record 5 edges · misfit 0.14 ft · area +0.06%
59 items on the exception queue 15 wrong line · 15 no line · the rest genuine or matchline	13 / 19 arc lengths matching the drawn curve was 4/11 · loop 2 · compound curves now split	1 strip tunnel easements 61985-1..4 as drawn one envelope, not four figures · settled, not blocked	6 / 6 sheets georeferenced from their own callouts two tiles · package check ≤ 0.4 ft, offset on none
6 / 6 sheets with LiDAR under them third tile from USGS 3DEP (2018) for the gap · 9 sheets on the bench			

Where we are

Two weeks of spikes proved the pieces. The plan reset on 09-22 named the wall: not reading, but **association**, which drawn line a printed number describes. The 09-23 overnight loop attacked that on the Presidio set with the CAD structure hidden in the PDFs (layers and AutoCAD SHX text annotations) and roughly doubled the pass rates. A second loop the same afternoon made the bench deterministic, tripled the arc-length passes by splitting the long R/W curves where the drawing marks a boundary, and settled the tunnel-easement question: the sheet draws them as one strip, so the four are queued by design, not by failure. What remains before 10-05 is polish and the run of show.

A third loop the same evening (18:00 to 21:10) took the test set from three sheets to six: the Presidio pair plus three sheets on the Marin approach of the Golden Gate Bridge (R-10741.1/.2/.3, 2017, Route 101, drafter CHaldenwang, a Civil 3D 2012-style export with no SHX annotations and no tables, read by OCR). All six are now georeferenced from their own printed callouts and placed on their tiles; Caltrans' packages check all six and are the offset on none. New tile plumbing (`tiles.py`) routes the LiDAR tail per sheet, `demo.py --six` runs the other five sheets through the chain (640 s, 0 failures), and `qgis_project.py --six` builds a two-tile, six-sheet project. Fixing two state bugs made the baseline reproducible: the closing row is `loop3-final` . Coverage: LiDAR lies under four of the six sheets; the bridge and the strait, where the tiles do not meet, need USGS 3DEP. Leg 5 attributed the north misses from 12 crops: the drafter writes the curve delta as `A=` and OCR joins angle and length into one block, so no curve on the north sheets was ever checked. Both are read now: curve $L=R\cdot\Delta$ passes 2 of 2 on R-10741.2 (exact), wrong-line 5 to 3 on .2 and 2 to 1 on .3, no other sheet moved. What is left there has no drawn line to check: chord citations beside curves and a not-to-scale detail inset.

A fourth loop (21:20 to 00:40) went after coverage. A third LiDAR tile from USGS 3DEP (2018, about 10 pts/m²) now lies under R-10741.1 and the east edge of R-10434.3, so every baseline sheet has ground beneath it; R-10434.3 itself turned out to sit 89.5 % on the Presidio tile once its fit was redone, so loop 3's "gap" claim for it was a stale fit. Two more annotation-bearing sheets place from their own control (R-71.71 by grid lines, R-10258.1 by callouts and grid, package agreeing to 0.09 ft), and a third, R-71.70, now takes NGS first-order marks as control: both PIDs fetched, both brass disks found by leader, but the two-point fit disagrees with the sheet's own bearings by 1.29°, so it stays unverified. On the six: the north drafter's dashes were 8.6 to 8.9 pt and the 8 pt cap had been chaining tick marks instead, so the dashed parcel line beside S16°20'26"E now carries its bearing (20/20 on R-10741.2); R-10434.3's line table was masked short by an unrelated curve table at the same x, so 12 table cells had been

checked as labels (wrong-line 45 to 33, every pass kept); comma-grouped distances read. One regression caught by the canonical bench and reverted before it shipped: a sliver guard added for R-71.70 had rerouted 462 real annotations on R-10434.3.

A fifth loop (09-24 evening) went after the south bearings, then the arcs. The bearings had one root cause: a chained line kept its first segment's direction and handed it to every piece cut from it, so every label along a long multi-piece record line was compared with one arbitrary heading. Each piece now takes the direction fitted through its own points, and the three south sheets went from 30/49, 18/31, 32/51 to 34/49, 20/31, 37/52 with nothing lost. The arcs did not move, and the attribution says why: the compound-curve story from loop 2 is down to 8 of 65 misses; 32 are bare distances a length-blind fallback was scoring as arcs (now gated, 7 fake fails gone), and 25 are standalone L= labels with no drawn arc within reach, the next thing to look at.

A sixth loop the same night took the six accuracy items in order. Arcs: a lone L= label now sums the contiguous pieces of the one curve it sits on, which brought back the 573.93 (T) total as 52.22 + 20.63 + 501.10 and five more (+6). Bearings: a tolerance that scales with the drawn piece's length, capped at 30 arcminutes, with pieces under 10 pt queued as too short to carry a bearing at all (+10, and five earlier passes on 5 to 8 pt pieces retired as luck); a bearing beside a curve is checked as the chord between the arc's ends (+4); a parallel neighbour is chosen by the printed bearing (+1); at a busy vertex the leader's landing piece wins (+1). Not-to-scale detail bubbles are now found by their dashed outline, caption or no caption: four on R-10434.1, one on R-10741.2, and four passes that sat inside them retired. Two hypotheses died honestly: the north drafter's font is not the read limit (the open simplex and romans strokes are identical to the seed already in use; the unread blocks are prose paragraphs), and R-10434.1's lost closed chain was a transient state that never closed for real. Bearing denominators now count checkable pieces only.

A seventh loop, the next morning, tried the scan idea from gap 1 on R-65.2: cluster the glyph images of one hand, name each cluster once, read the rest by template. The clustering is clean (98.6 % purity on the keyed glyphs once glyphs are separated) and the naming works (57 of 61 agreed boxes vote, the vision model names 36 clusters from their contact strips with no disagreement), but the reader lands at 7 right, 0 wrong, 25 queued on the keyed 32 against the two-reader consensus's 16, 1, 15. As a third vote it lifts the union to 18. What stops it is not recognition: hand digits touch and fuse in 197 of 273 boxes, hatch runs through text, and 8 of the 32 keyed boxes are detector boxes sitting on the wrong text before any reader runs. The next lever on scans is text-line detection on the binarised page.

An eighth loop, one leg, came from looking at one crop: the compound R/W curve on Presidio (R=1470.00', L=573.93'(T)) whose record is the table rows C15 and C16 with no drawn boundary between them. The run-sum rule existed for inline labels and for the small fillets, but the table-tag path grouped tags by the piece their leader landed on, and C15's leader lands on a radial tick. The tag path now uses the same parent-tagged arc pool as the inline checks and searches a tag's own curve for a window whose length matches: C16 passes as its two pieces (521.73 vs 521.76), C19 passes, C19's false radius fail is gone, and C21 is now measured on the right piece, where it still fails for real. Presidio row checks went 31 pass / 12 fail to 33 / 9 with nothing else moving on any sheet. C15 itself cannot pass alone: its drawn piece is C15 plus an untagged C18, and the (T) total that covers them already passes.

A ninth loop, one leg, followed the same crop. The thin alignment curve C21 points at is drawn as several PDF paths end to end, so curve paths whose ends meet with matching tangent and radius are now one parent; dash-dot patterns chain into trains like dashes; and a table's consecutive rows are checked as one run against the drawn run between the marks that exist, even when a row has no tag on the drawing. That last rule is what C15 needed: it passes as C15 + C16 + C18, the untagged neighbour, 573.95 against 573.93. C20 now reaches a check and fails honestly, since its leader lands on the alignment centreline. C21 is measured rather than fixed: its stitched parent reaches 706.79 of the record's 876.88 ft, and the next path, 10.65 pt away, is a different curve. Two things were tried and reverted with numbers: a wider join gained nothing on C21 and moved a neighbour's radius across tolerance, and dash-dot pieces reached by proximity alone produced six fake fails on R-10434.3, so they now need a leader.

A tenth, short loop traced C21 by eye with the checker's own candidate pool drawn on the sheet. It is all there: a tangent, then five pieces of one thin line across the C9/C10 fillet ($255.91 + 193.93 + 86.86 + 278.14 + 62.04 = 876.88$ ft), ending where three straight lines start along the curve's tangent, a point of tangency. Two rules stop the checker from seeing that: the join across the fillet compares a blended whole-piece radius (2.8 % off), and there is no point-of-tangency cut. Both fixes were tried and rejected on the no-loss gate, with numbers: radius fits on 40 ft windows swing from 1535 to 1382 ft at this sheet's two-decimal precision (100 ft is the floor), and a sheet-wide tangency cut fragments other curves and costs three passes elsewhere. The diff is kept; the scoped version is the next step. Nothing shipped; the loop 9 numbers stand.

An eleventh loop answered a question about the screen: why the purple road surface stops abruptly. Purple was the LiDAR vendor's bridge-deck class alone; it ends where the road leaves the structure, at the Battery tunnels (no return at all), where the class thins at parapets and joints, and on the 2018 3DEP tile, whose pavement came out as 190 fragments under the polygon floor. The class audit found ground on class 2 and

deck on class 17 on every tile, so the fix was geometry and assembly, not classes: deck now grows from class-17 seeds into dark, flat cells two metres above the ground DTM; fragmented tiles get a wider closing; and one highway surface per tile dissolves pavement and deck inside the right-of-way faces, marked structure or at grade. The tunnel easements are drawn as record only, below ground. Gains in area are small and honest; the continuity on the screen comes from the union. Found on the way: the one-command demo has slipped to 245 s after the curve work of loops 8 and 9, the first item on the list.

09-09 to 09-21

Spikes

Text from strokes, OCR, georef from callouts and grid ticks, LiDAR Q1-Q5, scan census. 8 of 11 CAD sheets georeference.

09-22

Plan reset

Association is the wall. Product = the numbers walked as a traverse. Scans = QA story, not input. 13-day roadmap.

09-22

demo.py, objects, QGIS

One command, offline. Object record with provenance. Exception page. Figures and numbers from the run.

09-23 04:00-09:00

Overnight loop

SHX annotation reads, association filters as defaults, traverse edges paired, 1969 scans on the tile. 7 commits, every gate re-measured.

09-23 12:00-14:30

Loop 2

Twin annotations folded. Dash trains on the easement layer. Compound curves split. Arcs 4/11 to 13/19. Easement strip measured as one envelope.

09-23 18:00-21:10

Loop 3

Six zones as the baseline: north tile plumbed, R-10741.1/.2/.3 georeferenced from their own callouts, two-tile QGIS project, bench row loop3-final; north curve data read, wrong-line down.

09-23 21:20-00:40

Loop 4

Coverage: 3DEP gap tile, ground under all six; R-71.71 and R-10258.1 placed; NGS PID control built (R-71.70 unverified, 1.29°); north dashes seen, R-10434.3 table cells masked; bench row loop4-final.

09-24 18:00-19:15

Loop 5

South bearings +11 from one cause (cut pieces inherited the seed direction); arcs re-attributed: fake arc fails gated out, fillets matched by radius; bench row loop5-final.

09-24 19:30-23:45

Loop 6

Lone L= arcs summed on their curve (+6); bearing tolerance by piece length, chords beside curves, parallels by printed

09-25 08:45-09:25

Loop 7

The scan by its own alphabet on R-65.2: clean clusters, thin coverage; a third reader (union 18 of 32, 0 wrong); detector

bearing (+16); detail bubbles by geometry; font-seed and closure hypotheses falsified; bench row loop6-final.

boxes on the wrong text are the larger loss.

09-25 11:10–17:10

Loop 8

Curve table tags on the shared parent-tagged arc pool; Presidio row checks 31/12 to 33/9; C15 shown to be a C15+C18 piece; bench row loop8-final.

09-25 17:40–20:15

Loop 9

Curve paths stitched into one parent; dash-dot trains; consecutive table rows as a run (C15 passes); C21 measured, not fixed; bench row loop9-final.

09-26 00:40–03:10

Loop 10

C21 traced by eye: whole on the sheet, five pieces to a point of tangency; two fixes measured and rejected; nothing shipped.

09-26 03:30–04:40

Loop 11

The road as one surface: class audit per tile, deck by height above ground, gap-tile pavement, tunnels record-only; demo --fast regressed to 245 s.

09-23 now

Day 13

Slides, Q&A sheet, recording not started. Remaining arc misses need a run-sum against a neighbour tag.

to 10-05

Freeze, rehearse

Fallback recording. Two rehearsals. Freeze by 10-03.

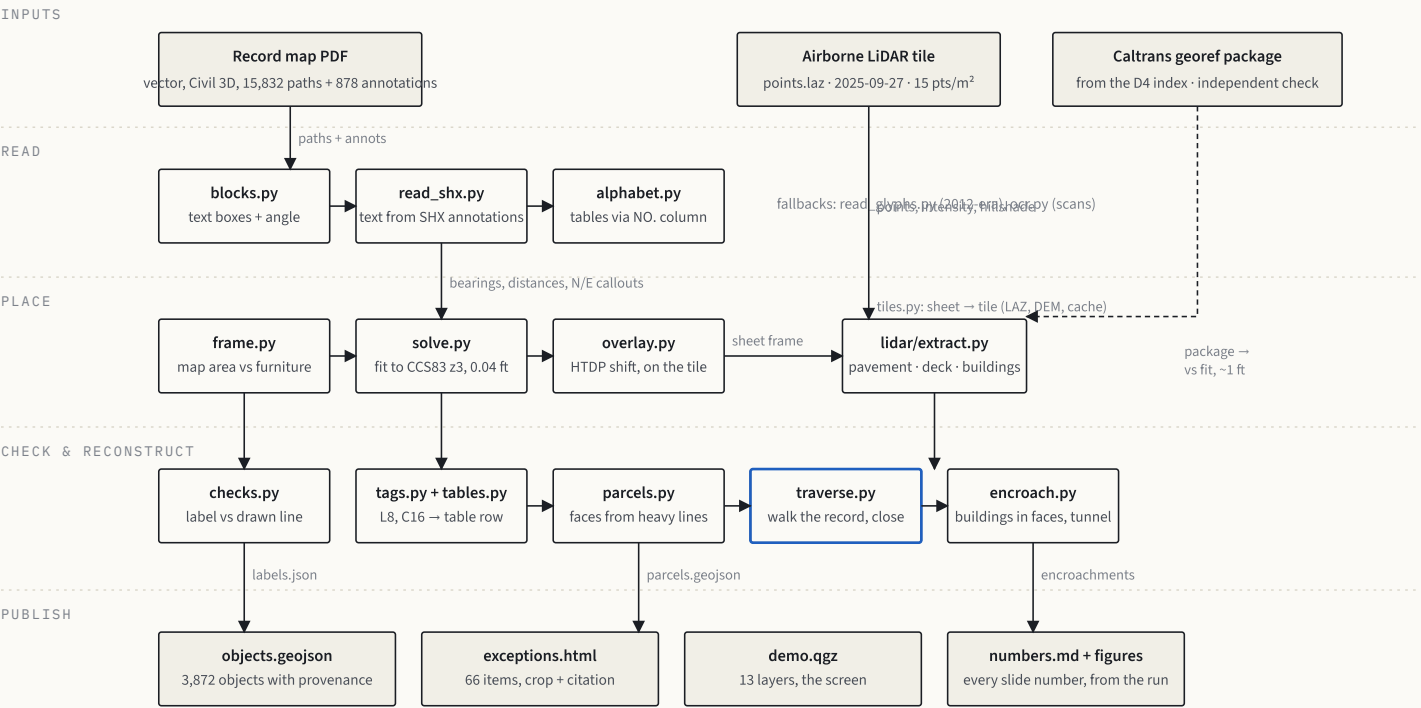
Roadmap items

DAYS	DELIVERABLE	STATE	PROOF
1–2	One command, offline, under 3 min. Object record with provenance. QGIS project.	DONE	122 s, 21 steps ok; 3,872 objects each with sheet, rule, residual, status; 13 QGIS layers valid, render PNG as proof
3–6	Table traverse: tags on the drawing tied to table rows, figures walked and closed	DONE AS FAR AS THE SHEET ALLOWS	tags associated 24 of 44 rows, line checks pass; 2 chains close on the demo sheet; the easements are one drawn strip, measured and queued as such (below)
7–8	Exception page a surveyor can work	DONE	one offline HTML, 66 items in cause groups, crop + citation + resolution field per item; scan's queue on the same page
9–10	LiDAR numbers and figures refreshed from	DONE	numbers.md and four figures regenerate at the end of every

	the run		run; tunnel profile caveated until easements close
11	Slides for the non-demo minutes, Q&A sheet, recorded fallback	NOT STARTED	Q&A answers drafted in ROADMAP.md; no recording yet
12-13	Two rehearsals, freeze	NOT STARTED	—
—	Six-sheet baseline (JR, 09-23 evening)	DONE	six.qgz 27 + 21 layers, bench row loop3-final, 5 dispatches

The demo workflow

Everything below runs from `python spike/demo.py --fast` with no network, 129 s. A `--six` flag adds the other five sheets' tails and the six-sheet QGIS project, 640 s, 0 failures. Twenty-one scripts in fixed order, each reading the files the previous ones wrote under `spike/out/`. Read once, then every downstream number is computed from geometry, never re-read.



The pipeline as it runs. Text is read once from the CAD annotations, then every check, closure and area is computed from geometry. The traverse step is the demo's thesis: the printed record

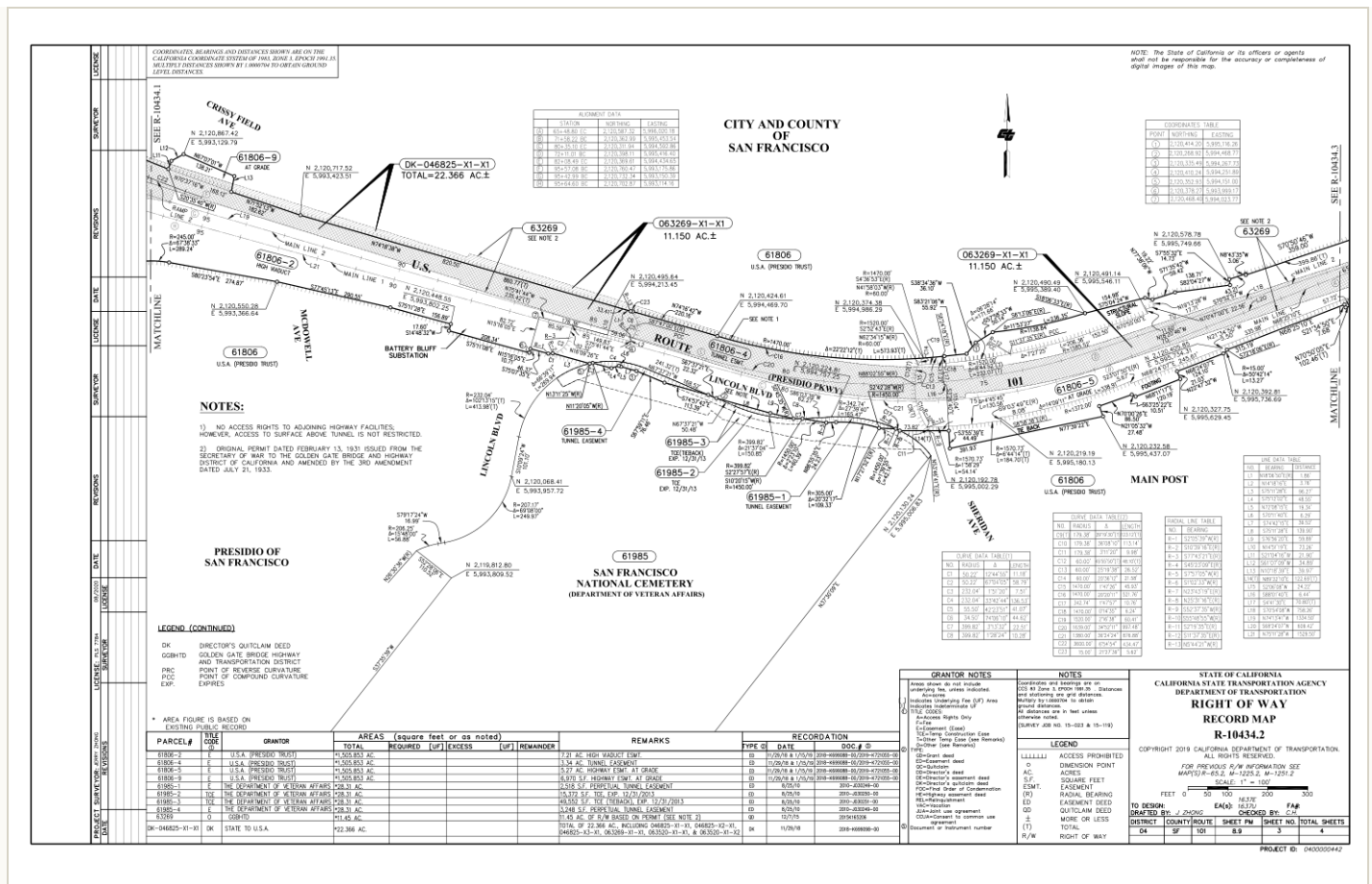
walked and closed with no label-to-line heuristic. Caltrans' own package enters only as an independent check on the fit.

The five demo segments

#	SEGMENT	WHAT THE ROOM SEES	STATE
1	Sheet in, objects out	Tables and callouts read; fit residuals; linework on the LiDAR intensity image	LIVE
2	The record checks itself	Tables walked as a traverse; closure per figure; printed vs drawn; exception queue with crops	LIVE
3	Record against ground	Parcels on LiDAR; pavement, deck, building extraction; what lies outside the R/W; HTDP before/after	LIVE
4	QA that catches its own mistakes	Two chance scan fits refused on the grid-label scale check; flightline QA	PAGE EXISTS
5	The scan, honestly	1969 sheets on the tile via Caltrans' package; 273 boxes, half readable, queue a surveyor would work	SLIDE

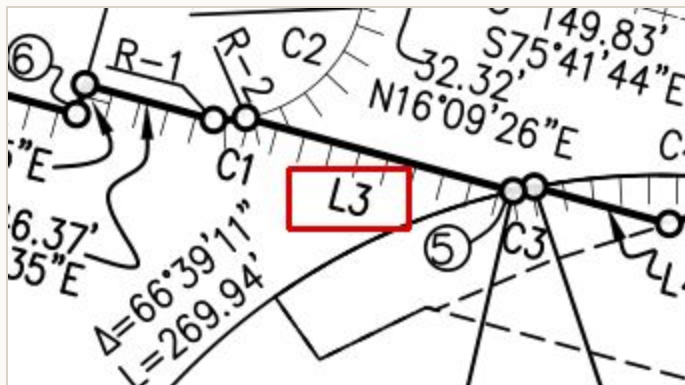
What the workflow does, on one tag

Follow segment tag **L8** from the drawing to a verdict. This is the core loop for every table row, every inline label and every traverse edge.



LINE DATA TABLE		
NO.	BEARING	DISTANCE
L1	N18°06'50"E(R)	1.86'
L2	N14°18'16"E	3.76'
L3	S75°11'28"E	96.27'
L4	S75°12'02"E	48.55'
L5	N72°08'15"E	19.34'
L6	S70°11'40"E	6.29'
L7	S74°42'15"E	39.52'
L8	S75°11'28"E	139.90'
L9	S76°56'20"E	59.89'

2. Read the row. The NO. column is a known sequence, so the table validates itself. L8 says S75°11'28"E, 139.90 ft.



3. Or queue it. L3 has no leader and two candidates. Ambiguity goes to the queue, never a guess.

READ_SHX.JSON · ONE BLOCK PER ANNOTATION

```
{ "cx": 1837.5, "cy": 757.0, "w": 36.0, "h": 7.26,
  "angle": -12.76, "text": "TIE BACK",
  "conf": 1.0, "src": "shx", "id": 2 }
```

TABLES.JSON · L8 AFTER ALPHABET.PY

```
L8    S75°11'28"E    139.90'
```



TAGS_CHECKS.CSV · VERDICT PER ROW

tag	check	printed	drawn	diff	result	assoc
L3	bearing	S75°11'28"E	S75°10'36"E	0.9'	pass	beside
L3	distance	96.27	96.27	-0.00	pass	beside
L4	bearing	S75°12'02"E	S75°12'43"E	0.7'	pass	leader
L4	distance	48.55	48.56	+0.01	pass	leader

GEOREF.JSON · THE FIT EVERY CHECK LIVES IN

```
"crs": "EPSG:2227", "scale_ft_per_pt": 1.38885,
"rotation_deg": 0.536, "rms_ft": 0.041,
"control": [{ "N": 2119812.80, "E": 5993809.52,
               "residual_ft": 0.053, "used": true }, ...]
```

 Twelve random passing distance labels, each with the label in a red box and the measured line in blue

Output: twelve random passing distance labels, label region in red, the line the reader measured in blue. This is the gate evidence the overnight loop used: a pass only counts if a human sees it on the right line. All twelve are.

Then the record walks itself

TRAVERSE.JSON · ONE CHAIN, CLOSED

```
{ "closed": true, "n_edges": 5, "full_record": 5,
  "misfit_end_ft": 0.14,
  "edges": [
    { "edge": "N67°07'01\"W + 138.21'", "misfit_ft": 0.07 },
    { "edge": "L12", "misfit_ft": 0.11 },
    { "edge": "L11", "misfit_ft": 0.07 },
    { "edge": "N70°37'16\"W + 168.12'", "misfit_ft": 0.10 },
    { "edge": "L13", ... } ] }
record area 6,970.6 sq ft · drawn face 6,974 · +0.06 %
```



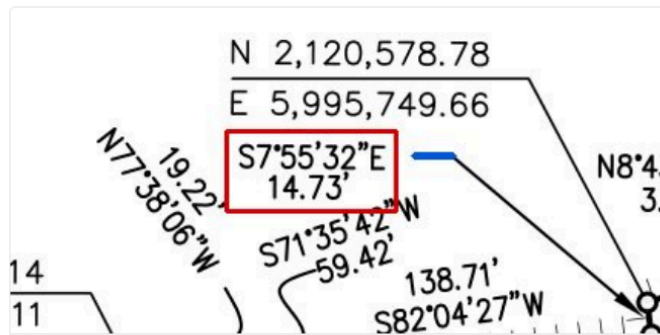
OBJECTS.GEOJSON · ONE OF 3,872 PUBLISHED OBJECTS

```
{
  "kind": "control point",
  "sheet": "r_10434_002_2020-09-16",
  "printed": {
    "E": 5993809.52,
    "N": 2119812.80
  },
  "measured": {
    "sheet_pt": [797.8, 1026.4]
  },
  "residual": 0.053,
  "units": "ft",
  "status": "verified",
  "rule": "printed N/E callout, leader traced to point,
    similarity fit by consensus; residual vs fit",
  "adjudicated_by": "",
  "adjudicated_on": ""
}
geometry: Point [-122.46557, 37.80025]
```

Every published object carries the sheet, the printed value, the measured value, the residual, the rule that produced it, and an empty adjudication slot. In QGIS a click shows all of it. That is the "reviewer reaches provenance in seconds" promise.

▼ Printed value does not match the line beside it 7

A dimension or bearing disagrees with the line the reader measured (blue) by a small amount. Decide which is the record.



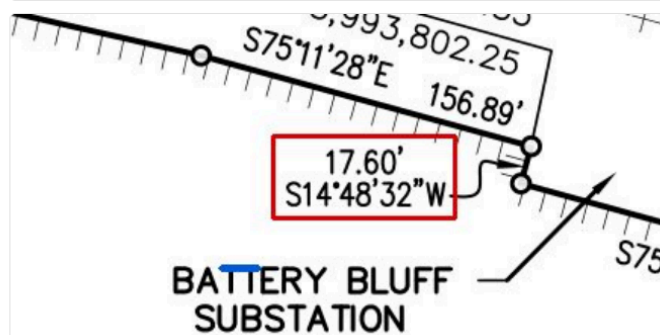
distance

distance: off by +2.60 ft

printed 14.73' · drawn 17.33 ft · +2.60 ft

r_10434_002_2020-09-16 · region 2063, 432, 2118, 458 pt

— resolution — initials / note



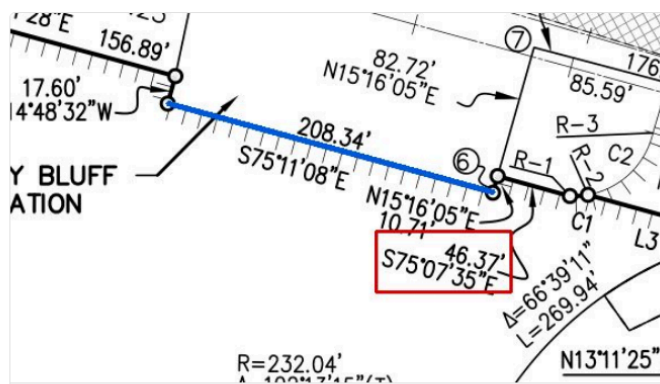
distance

distance: off by -1.35 ft

printed 17.60' · drawn 16.25 ft · -1.35 ft

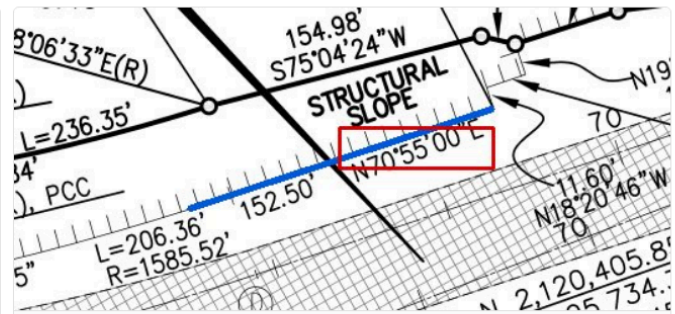
r_10434_002_2020-09-16 · region 712, 565, 772, 592 pt

— resolution — initials / note



bearing

bearing: off by +3.4'



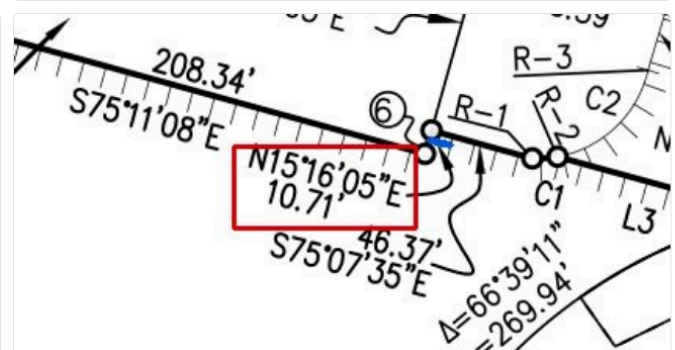
bearing

bearing: off by +18.6'

printed N70°55'00"E · drawn N71°13'36"E · +18.6'

r_10434_002_2020-09-16 · region 1992, 558, 2053, 574 pt

— resolution — initials / note



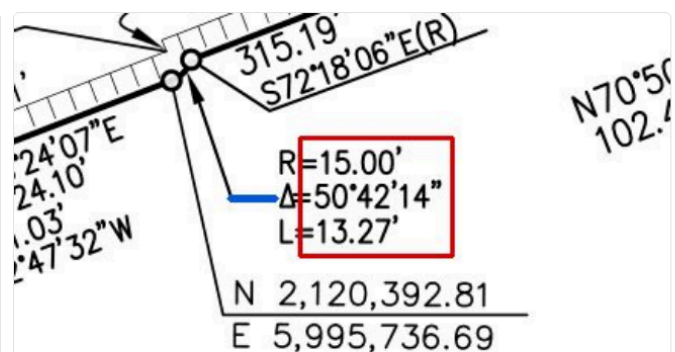
distance

distance: off by -0.75 ft

printed 10.71' · drawn 9.96 ft · -0.75 ft

r_10434_002_2020-09-16 · region 877, 618, 935, 644 pt

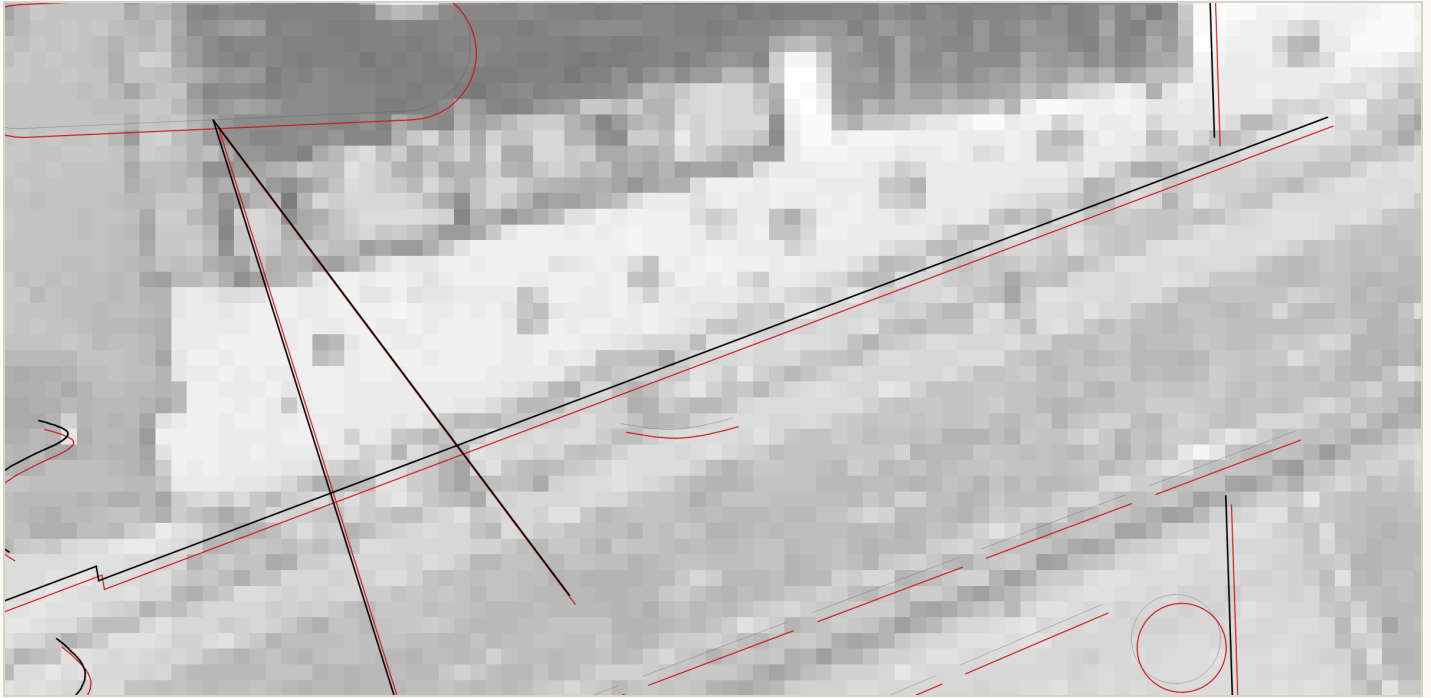
— resolution — initials / note



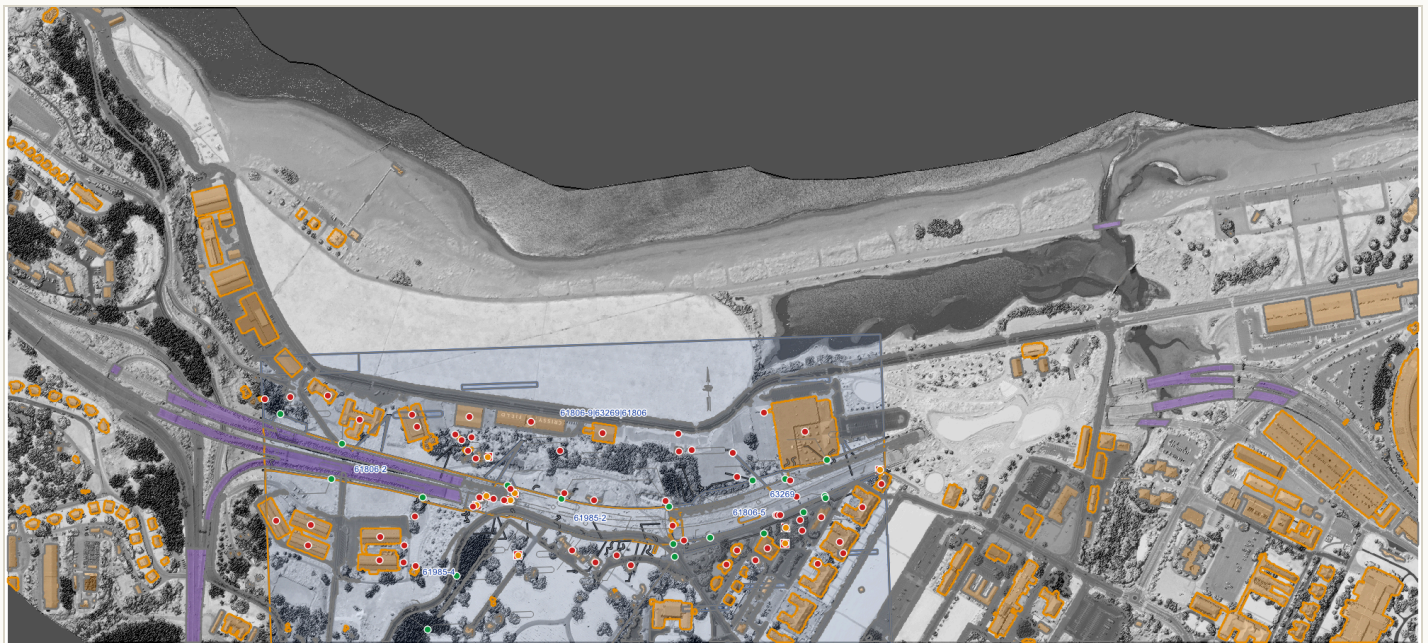
distance

distance: off by +4.58 ft

Output: exceptions.html. One offline file. Each item: crop, cause group, printed vs drawn, citation in PDF points, a resolution a surveyor fills in and exports.



Output: linework on the LiDAR, before and after HTDP. Sheet epoch 1991.35 to LiDAR 2010.0:
0.668 m toward N33°W, applied and shown, never silent.



Output: demo.qgz, the screen. Thirteen layers built by PyQGIS in legend order. The script re-reads the written project, asserts every layer survives, and renders this PNG as proof.

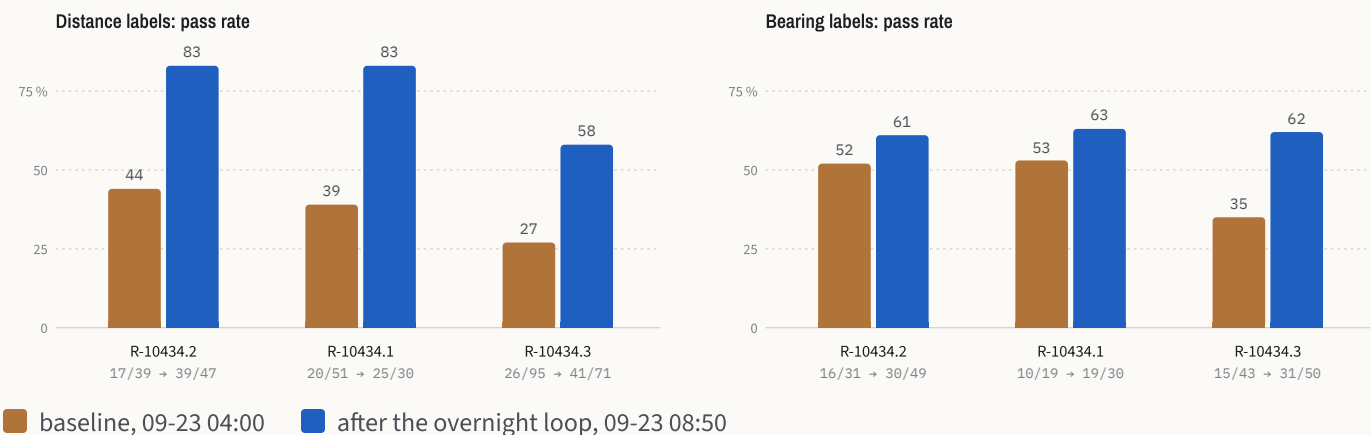
Accuracy on the nine-sheet bench (six-sheet baseline)

Six sheets, two LiDAR tiles: three Presidio vector sheets by one drafter (R-10434.1/.2/.3, 2020) and three on the Marin approach of the Golden Gate Bridge by another (R-10741.1/.2/.3, 2017, Route 101, no SHX annotations, read by OCR). Every number re-measured by `spike/bench.py loop3-final` on 09-23 at 20:05, not from memory; the row was written twice and matched. R-10434.1 and .3 move from the loop 2 prose (25/32, 41/70 distance passes) to 26/37, 42/70 here because those loop 2 rows were measured on stale sibling state, fixed in loop 3 leg 3; Presidio is unchanged. "Pass" means the printed value agrees with the drawn geometry under the fit: distances to a few hundredths of a foot, bearings to under a minute, on both tiles.

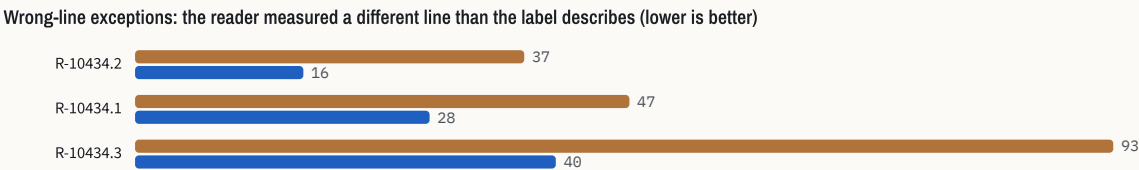
SHEET	TILE	BLOCKS READ	DISTANCE	BEARING	ARC	WRONG - LINE	GEOREF RMS	PACKA
R-10434.2 (Presidio)	South	831/831	41/49	41/43	16/19	12	0.04 ft	~0.1.1
R-10434.1	South	721/721	26/37	21/23	12/24	21	0.02 ft	n
R-10434.3	South + gap	1403/1403	42/59	41/42	26/39	27	0.02 ft	n
R-10741.1	Gap (3DEP)	311/311	14/18	13/13	1/2	3	0.04 ft	0.29 (9 pt ma
R-10741.2	North	381/382	20/20	18/18	0/1	0	0.05 ft	0.37 (9 pt ma
R-10741.3	North	348/349	17/17	13/13	0/1	1	0.03 ft	0.40 (9 pt ma
R-71.71 (2024, SHX)	none on disk	209/209	14/19	20/20	1/5	8	0.01 ft (13 grid lines)	packa on di
R-10258.1 (2020, SHX, metric)	none on disk	434/434	12/49	22/23	0/18	36	0.00 ft (6 callouts + 11 grid lines)	0.09 (9 pt ma

R-71.70 (2019, SHX, zone 2)	none on disk	—	0/5	0/2	0/8	15	unverified: 2 NGS marks, bearings 1.29° off	packa on di
--------------------------------------	--------------------	---	-----	-----	-----	----	---	----------------

N/M = NOT MEASURED AGAINST THE PACKAGE FOR THAT SHEET. NORTH "VS PACKAGE" IS THE MAX RESIDUAL AT 9 SHARED SHEET POINTS; SOUTH IS PRESIDIO'S FIGURE FROM THE SCORECARD BELOW, RASTER-LIMITED.



Distances roughly doubled, bearings up 10 to 27 points. Denominators changed because reads changed: annotations find more labels, and stationing, table cells and acreages no longer count as labels. Fractions under each pair.



Wrong-line, the weakest link, more than halved. On the demo sheet it went 103 (first exception page) to 16. Fixes: leaders by arrowhead, stubs and underlines excluded, runs split at circles and junctions, printed bearing and span as filters.

Full scorecard, R-10434.2 (demo sheet)

MEASURE	RESULT	NOTE
Reading		
Table rows read clean from SHX annotations	44 / 44	R-10434.1 25/25, R-10434.3 83/83. Hand-keyed cells 111/111 match
Text blocks with exact text	832	annotation text, confidence 1.0; no OCR on this sheet class

Georeferencing

Control points used, RMS	18 / 19 · 0.04 ft	derived scale 1.38885 ft/pt vs plot scale 1.38889
--------------------------	-------------------	---

Agreement with Caltrans' own package	~0.6–1.1 ft	raster-limited; the independent check
--------------------------------------	-------------	---------------------------------------

Printed vs drawn (labels on the drawing)

Distance	41 / 49	passes agree to 0.03 ft median, 0.24 max
----------	---------	--

Bearing	41 / 43	passes agree to 0.55' median
---------	---------	------------------------------

Arc length	16 / 19	was 4/11. Long R/W curves now split at circles, meeting lines and radial ticks; (T) totals checked. Misses left: boundaries the drawing does not mark
------------	---------	---

Curve $L = R \cdot \Delta$ (internal consistency)	10 / 10	
---	---------	--

Table rows through their drawing tags

Rows found on the drawing / associated	33 of 44 / 24	18 by leader arrowhead; 12 partial reads and 33 ambiguities queued, none guessed
--	---------------	--

Row checks pass · fail	34 · 10	where associated, lines are exact; the arc fails are runs whose sub-arc boundary is not drawn
------------------------	---------	---

Reconstruction

Traverse edges placed / with full record	87 / 49	was 81 / 22 before the loop
--	---------	-----------------------------

Closed chains	2	5 edges, misfit 0.14 ft, area 6,970.6 vs drawn 6,974 sq ft; a second 4-edge loop closes to 0.18 ft
---------------	---	--

Parcel faces named / within 1 % of the parcel table	8 of 60 / 1	figures leave at matchlines; easement strips do not close
---	-------------	---

Tunnel easements 61985-1..4	1 strip · 48,410 sq ft	drawn as one envelope; table sum 70,690 sq ft, the TCEs overlap the tunnel easements. QUEUED BY DESIGN
-----------------------------	------------------------	--

LiDAR

Features extracted	35 · 11 · 281	pavement, viaduct deck, building polygons (175 pass the roof test); deck now also grown by height above ground (+45 m ² inside the R/W)
--------------------	---------------	--

Highway surface, Presidio tile	33,988 · 146,620 m ²	structure · at grade, one dissolved surface inside the R/W faces; the blank middle is the Battery tunnels, drawn record-only
Highway pavement inside drawn R/W faces	81 %	9,014 m ² outside within 15 m flagged for review
Epoch shift 1991.35 → 2010.0	0.668 m N33°W	NGS HTDP v3.6.0, real call
Beyond the demo set		
Six-sheet baseline (all six sheets, two tiles)	↑ table above	distance, bearing, arc, wrong-line, georef rms, vs package per sheet — see “Accuracy on the six-sheet baseline”
Vector sheets georeferenced (all D4 vector sheets on disk)	7 / 17	the 10 refused all fail on reads: other drafters' fonts, no tables to validate an alphabet (before the R-10741 set)
Scan R-65.2 (1969, hand-lettered), 32 keyed boxes	18 right · 0 wrong (gated) · 14 queued	three readers: RapidOCR, a local vision model, and the sheet's own alphabet (loop 7); 8 of the 14 are detector boxes on the wrong text. 48 of 65 sheets in the archive are scans

What the numbers do not claim. The 0.04 ft residual is against numbers read off the same sheet. Independent check is Caltrans' package at about a foot, raster-limited. We claim records-research grade, not a survey tolerance, until a pilot sets one.

The easements, settled

Tunnel easements 61985-1..4 are one drawn strip, not four figures. Loop 2 chained the 345 single-dash strokes on the `rw_EASE_EXIST_align` layer into eight trains. The strip has a north edge drawn under the R/W line, a south edge parallel to it, and closures at both ends. No stroke on the sheet divides it into four. The parcel bubbles' leaders go to R/W vertex circles, and the printed curve data sits on the R/W line itself.

Measured: the envelope is 48,410 sq ft against a table sum of 70,690. The two temporary construction easements overlap the two tunnel easements rather than tiling the strip. Their individual outlines live in the deeds, not on this sheet.

For the demo: segment 2 shows the two chains that close, the strip envelope with its area, and the four easements queued with this reason. That is the story: the system

measures what the sheet draws and says what it cannot prove.

 Strip region of the sheet with the dashed easement trains coloured, the envelope filled, and the four parcel ovals labelled with their table areas

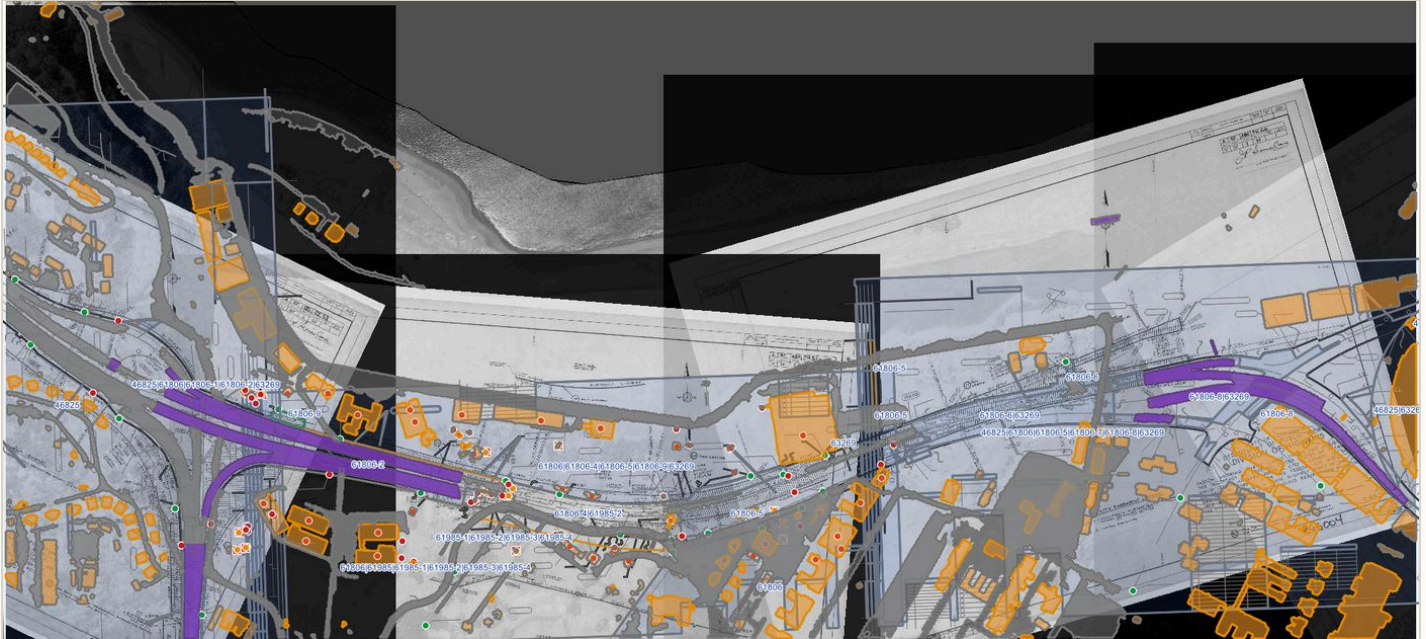
The strip as drawn. Dash trains from the easement layer, the north edge in blue under the R/W line, the envelope filled, the four ovals with their table areas.

The north set

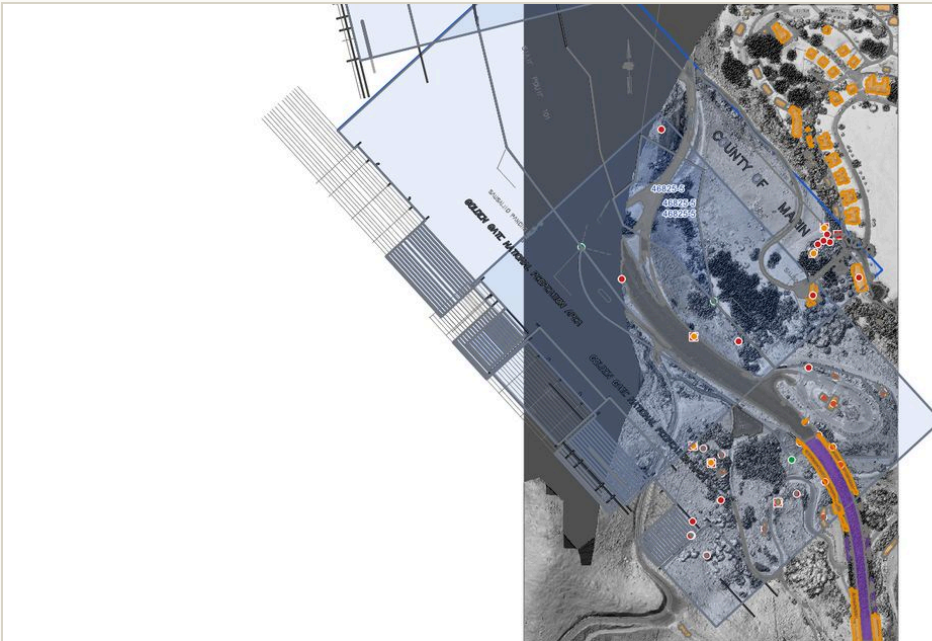
CHaldenwang's callouts read differently from the Presidio drafter's. Coordinates print as a two-row block, N over a bar over E, with a thin (< 1.0 pt) leader running to a filled arrowhead that sits 10.6–10.9 pt short of a small circle rather than touching it. Civil 3D 2012-style export: no SHX text annotations, no tables, so every block is read by OCR. Three callouts paired on R-10741.1, four on .2, three on .3 (10 of 10 traced) once the leader trace learned two rules: walk across thin weights (below 1.0 pt) instead of stopping at them, and snap to a circle within 12 pt of the arrowhead when no second circle sits within 24 pt.

Placement: sheet-derived fits (scale and rotation from the sheet's own bearings and distances, offset from the traced callouts) land at rms 0.04 / 0.05 / 0.03 ft on .1/.2/.3. Checked against Caltrans' package at 9 sheet points: max 0.29 / 0.37 / 0.40 ft. .2 is credible outright; .1 and .3 are weaker fits (3 callouts each) with the rotation confirmed independently by 12 printed bearings apiece. Sheet-derived scale agrees with the package to 0.011 % and rotation to 0.0051° . The package is now the check on all six sheets and the offset on none.

Coverage: the south tile covers R-10434.1 and .2 only — R-10434.3 is 0 % inside it, its northing starting 5 m past the tile edge (STATE.md's earlier "7 sheets" claim was wrong). The north tile covers R-10741.3 (99.3 %), .2 (59.9 %), .1 (0.3 %). LiDAR lies under four of the six sheets; the bridge and the strait, where neither tile reaches, need USGS 3DEP.



South tile, six.qgz. R-10434.1/.2/.3 linework, parcels and the 1969 record group over the Presidio DEM hillshade and intensity, 27 layers.



North tile, six.qgz. R-10741.1/.2/.3 linework and parcels over the Marin DEM hillshade and intensity, 21 layers.

Next steps

In order. Loop 3 placed all six sheets and dropped leg 5 (north label association) when JR ended the loop, so it leads the list. Items 1 to 5 are code or a package fetch, gated against `bench.py`. Item 6 is the demo itself.

01 **demo.py --fast back under 180 s**

Now 245 s: checks 64 s, tags 51 s, tables 66 s after the curve stitching, dash-dot trains and window searches of loops 8 and 9. Profile the three and restore the gate without moving the loop 9 numbers. Then a highway surface for the gap tile.

02 **Scan text-line detection on the binarised page**

Replace the OCR detector's boxes with one box per lettered line built from glyph components; gate on the 32 keyed boxes landing on their text (today 24 of 32). Then the alphabet reader and the two generic readers vote per box.

03 **Row segmentation of multi-line blocks on 2012-era sheets**

R-10741.2's unread blocks are GRANTOR NOTES and TITLE CODES paragraphs; the font is not the limit. Reads first, then the OCR-vs-glyph selector re-measured.

04 **South bearings, what is left**

2 / 2 / 1 fails on the three sheets plus the too-short queue (6 / 8 / 11 pieces under 10 pt). The queue is the drawing's own limit; the fails need crops.

05 **C21 and the point-of-tangency class, scoped**

Traced: C21 is five pieces across the C9/C10 fillet ending at a point of tangency. Fix, scoped: radius fits on windows of at least 100 ft, and a tangency cut applied only inside the parent of a tag whose own window search failed, never sheet-wide. Diff at `spike/out/loop10_legA.patch`; gate as loop 10's.

06 **Arcs, what is left**

Presidio 3 fails, R-10434.1 12 (matchline cuts and details), R-10434.3 13. Classes (b) and (c) in `spike/out/leg6A_missing.md`.

07 **R-71.70's 1.29°**

Crop both NGS marks against the two-point fit; read the basis-of-bearings note. Then NGS control is a second independent check on every sheet that prints a PID.

08 **South + gap mosaic; the 2024 San Francisco tile**

R-10434.3 should keep the 2025 tile where it has it. CA_SanFrancisco_B23 covers both gap boxes but its LAZ carries no CRS.

09

The 1950s scans on the north tile

R-71.11/.20/.28 and R-92.8/.9 through their Caltrans packages the same way R-65.1/.3/.4 went onto the Presidio tile.

10

Slides, Q&A sheet, recorded fallback, rehearsals, freeze DAYS 11-13

Non-demo minutes and the scan segment as slides; every number from numbers.md. Record one full run. Two rehearsals inside 25 minutes. No code after the freeze on 10-03.

11

Parked until after 10-05

Scan reading by per-sheet template alphabet (gap 1, the archive's biggest cost). Per-object uncertainty. More drawn-geometry work on 61985-1..4: their outlines are in the deeds.

What has been built

Forty-eight Python scripts, about 8,000 lines, all under `spike/`. Throwaway by design: they answer "does this work" and feed the demo. Grouped by role; pipeline scripts in the order `demo.py` runs them.

FILE	WHAT IT DOES
Pipeline, in run order	
<code>demo.py</code>	Runs the 21 steps in ord offline (<code>--fast</code> , 129 s' and glyph reading wher annotation text covers t Routes QGIS steps to QC Python. <code>--six</code> adds tl five sheets' tails and the QGIS project (640 s, 0 fa
<code>blocks.py</code>	Text blocks from vector strokes: box and angle f piece of lettering, no OC
<code>read_shx.py</code>	Text from AutoCAD SHX annotations, one block annotation. The reader

	2016+ sheets. Confidence construction.
<code>frame.py</code>	Where the map is on the map area vs furniture (table block, legends), from the map itself.
<code>alphabet.py</code>	A sheet's own alphabet tables: the NO. column in the sequence, so its glyphs exemplars. Reads the linear and radial tables.
<code>read_glyphs.py</code>	Fallback reader for sheets without annotations: even if not matched against the sheet's alphabet. A glyph match
<code>ocr.py</code>	Fallback reader for scanned sheets: each crop, RapidOCR, plot notation.
<code>solve.py</code> + <code>georef.py</code>	One solver over every plot control type: N/E callout leaders traced to their plot monument notes, grid to Consensus fit to CCS83: axis redundancy, refuse
<code>tiles.py</code>	Maps a sheet to its LiDAR DEM, cache dir under <code>spike/lidar/cache</code> , the LiDAR tile steps taken from it instead of a hard Presidio path.
<code>overlay.py</code>	Linework to NAD83(2011 HTDP epoch shift, draw LiDAR intensity image.
<code>parcels.py</code>	Faces from the heavy linear named by parcel labels leaders, checked against table areas.
<code>checks.py</code>	Does what the sheet says what it draws? Each bearing

	distance and arc label a line under the fit; the ex queue.
tags.py + tables.py	Segment tags (L8, C16) drawing tied to table row leader arrowhead, else heavy line beside; two c go to the queue. Per-row
dashes.py	Chains the single-dash the easement layer into (dash to dash by headin deduplicated by covera; checks and parcels see lines as geometry.
traverse.py	The record twin: placed edges chained by share walked by bearing and closure and area where closes.
lidar/extract.py	Pavement, viaduct deck building polygons from returns, deterministical sheet's frame.
encroach.py	Building-class returns ir sheet's own R/W and ea faces; terrain profile ove tunnel easements.
export_rasters.py	Intensity ortho and hills GeoTIFFs for QGIS.
objects.py	The object record: every thing with sheet, printed measured values, resid status, adjudication slot
exceptions_page.py	One offline HTML page: queued item with crop, printed vs drawn, citatic resolution field. Scan's the same page.

qgis_project.py	The screen: PyQGIS built project with every layer ordered, re-reads it, re-opening view as proof. builds two tile groups (2 layers) and renders one tile.
figures.py + slide_numbers.py	One figure per demo sequentially rendered from the project number the slides quoted in run's files.
Measurement and gates	
bench.py	The gate instrument: run chain on a sheet set and summarises pass, fail, with tags, faces, traverse, take sheet, plus frame / block bearings_parsed / distances_parsed over the sheet set. Every loop leg accepted or rejected on
gt.py, gt_scan.py	Hand-keyed truth: 227 taken on R-10434.2, self-checked R-Δ; 34 boxes on the 196
sheets.py, batch.py	Generalisation runs: the chain over every vector disk and over unrelated maps from the public in
leg4_crops.py, legB_arcs.py, leg5_61985.py, checks_debug.py	Gate evidence and diagram random passing distance and every passing arc re their lines; the four ease walked as chains and the envelope measured; the line per label drawn.
Scans (most of the archive)	
scan_read.py, scan_readers.py, scan_vlm.py	Detect text boxes on a 3 render; RapidOCR and a vision model (qwen2.5v

	against keyed truth. No values in any prompt.
scan_consensus.py, scan_georef.py	Accept a read only wher readers agree or structu it; fit callouts through th solver, refused unless sc matches the grid labels.
scan_on_tile.py	The four 1969 R-65 scan georeferenced by Caltra packages, HTDP-shifted LiDAR tile as the "1969 r QGIS group.

LiDAR questions answered once

lidar/q1..q5	Flight date from GPS tin 09-27); record on terrair epoch offset by real NG! encroachment; flightlin QA (RMSE 0.02–0.04 m c
--------------	--

Documents

STATE.md, ROADMAP.md, spike/LOOP.md	Resume point with next every known trap; the 1 and the Q&A sheet; the loop's legs, gates and oi
spike/out/loop_report.md, legC_misses.md, numbers.md, exceptions.html	The overnight report; th attribution; the slide nu queue.

Findings worth carrying forward. Annotation text is the reader for Civil 3D 2016+ sheets; the glyph alphabet stays for 2012-era sheets. CAD layers did not help association once reads were exact; they still separate furniture from map. The exception page's wrong-line group is the fastest diagnostic there is: render the measured line, look at twelve crops, fix the biggest cause, repeat. Two generic readers agreeing on hand lettering is a floor, not a method.

```
SPIKE/OUT/NUMBERS.MD, SPIKE/OUT/BENCH.CSV (ROWS "LOOP9-  
FINAL", UNCHANGED BY LOOP 10; SCAN NUMBERS FROM  
SPIKE/GT_SCAN.PY AGAINST READ_SCAN.JSON AND  
READ_ALPHABET.JSON) · EVERY NUMBER RE-MEASURABLE WITH  
PYTHON SPIKE/BENCH.PY FINAL --TABLES --TAGS --PARCELS --TRAVERSE  
PRESIDIO R10434_1 R10434_3 R10741_1 R10741_2 R10741_3 R71_71  
R10258 R71_70
```